Answers and talk notes

Describe Data by Mean Absolute Deviation

What your student is learning

Tonight we use MAD — mean absolute deviation — to compare two groups that have the SAME average but do not behave the same at all.

Answers

Ask your student to explain before you look.

1. **A.** Scattered far from the mean. A large MAD means the values are scattered far from the mean. So the answer is: Scattered far from the mean.
2. **B.** Set B is more spread out than Set A, even with the same mean. Set B's values are 9 away from 10, while Set A's are 1 away, so Set B is more spread out than Set A, even with the same mean.
3. **C.** 2. The distances are 3, 1, 1, 3. Their sum is 8 and 8 ÷ 4 = 2.
4. **A.** 4. The distances are 6, 2, 2, 6. Their sum is 16 and 16 ÷ 4 = 4.
5. **B.** Set N varies more because its MAD is larger. A larger MAD means more variation, so Set N varies more because its MAD is larger.

Talk about it

The means match. How does the MAD show the difference?

Home mission

Two bus routes both average 20 minutes. Route 1 takes 19, 20, 20, 21. Route 2 takes 10, 15, 25, 30. Ask your student which route they want on a school morning, and why.

When they are stuck, ask

- How far is this value from the mean? Distances cannot be negative. A value equal to the mean has distance 0.
- Did you average the distances, or the original values?
- Same mean — does that mean the two sets look the same?

Try not to say

- Add the deviations with their minus signs, they will work out.
- If the means match, the data sets match.
- A MAD of 0 just means you did it fast.